

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

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ALGEBRA (Ph.D. Version)

Instructions to the student

- Answer all six questions; each will be assigned a grade from 0 to 10.
 - Use a different booklet for each question. Write the problem number and your **code number** (not your name) on the outside of the booklet.
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- Let G be a group acting on a set S . Assume that the action is transitive (that is, given $s, t \in S$, there exists $g \in G$ such that $gs = t$), and that the action is faithful (that is, if $g \in G$ satisfies $gs = s$ for all $s \in S$, then $g = 1$). Fix $s \in S$ and let G_s be the stabilizer of s . Show that if N is a normal subgroup of G such that $N \subseteq G_s$, then $N = 1$.
(b) Suppose G is a finite abelian group that acts faithfully and transitively on a set S . Show that G and S have the same number of elements.

- Let

$$M = \begin{pmatrix} 1 & 2 \\ 4 & 3 \end{pmatrix},$$

and let $R = \mathbb{C}[M]$ be the span over the complex numbers of $I, M, M^2, M^3, \dots$.

- Determine the dimension over $\mathbb{C}$ of R .
 - Find two non-zero matrices A and B in R such that $AB = 0$.
- Let p be prime, let $n \geq 1$. Let $R = (\mathbb{Z}/p^n\mathbb{Z})[X]/(X^p - 1)$.
 - Let M be the ideal of R generated by p and $X - 1$. Show that M is a maximal ideal of R . (Hint: Consider the homomorphism $f(X) \mapsto f(1) \pmod{p}$.)
 - Show that M is the unique prime ideal of R . (Hint: Show that $(X - 1)^p \equiv 0 \pmod{p}$.)
 - Let V and W be finite-dimensional vector spaces over $\mathbb{C}$. Let A be a linear transformation of V and let B be a linear transformation of W . Consider the map

$$A \otimes I - I \otimes B : V \otimes W \rightarrow V \otimes W$$

defined by mapping $v \otimes w$ to $(Av) \otimes w - v \otimes (Bw)$. Show that this map is invertible if and only if the set of eigenvalues of A is disjoint from the set of eigenvalues of B .

- Let K be a field of characteristic 0. Let n be a positive integer and assume that K contains the n th roots of unity. Let L/K be a finite Galois extension with Galois group G and assume that G is abelian of exponent n (that is, $g^n = 1$ for all $g \in G$). Let

$$B = \{x \in K^\times \mid \sqrt[n]{x} \in L\}.$$

- Let $x \in B$ and $g \in G$. Choose $y \in L$ such that $y^n = x$. Show that

$$\langle x, g \rangle \stackrel{\text{def}}{=} \frac{g(y)}{y}$$

is an n th root of unity that is independent of the choice of y as an n th root of x .

(b) Show that if $g, h \in G$ and $x \in B$, then $\langle x, gh \rangle = \langle x, g \rangle \langle x, h \rangle$.

(c) Let $x \in B$. Show that if $\langle x, g \rangle = 1$ for all $g \in G$, then x is the n th power of an element of $K^\times$.

(Note: You may not quote results from Kummer theory since they are what you are proving here.)

6. (a) Let G be a finite group and let A be a subgroup. Let $r_1, \dots, r_d$ be a set of coset representatives for G/A . Let $\rho : G \rightarrow GL(V)$ be a finite-dimensional representation of G over the complex numbers and let W be a subspace of V that is stable under the action of A (that is, $\rho(A) \cdot W \subseteq W$). Let

$$U = \sum_{i=1}^d \rho(r_i) W.$$

Show that U is stable under the action of G .

(b) Suppose that (ρ, V) is an irreducible representation of a finite group G and that G has an abelian subgroup A . Show that $\dim(V) \leq [G : A]$. (Hint: What are the irreducible representations of A ?)